BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

WORK INTEGRATED LEARNING PROGRAMMES (WILP)

Dissertation / Project / Project Work

Self-Correcting Financial Intelligence Systems

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AIMLCZG628T : DISSERTATION OUTLINE

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DISSERTATION / PROJECT / PROJECT WORK TITLE	**Self-Correcting Financial Intelligence Systems**

Contents

1. Broad Area of Work

The dissertation belongs to the discipline of Artificial Intelligence and Machine Learning, with primary focus areas in Natural Language Processing (NLP), Information Retrieval, Financial Document Understanding, and Evaluative MultiAgent Systems. The project aims to design and implement a selfcorrecting financial intelligence platform that unifies multiagent retrieval, teacher–student reasoning, and evaluator feedback for trustworthy automation in financial analytics and document comprehension.

Subdomains and technical focuses include:

1.RetrievalAugmented Generation (RAG): Implementation of a multiagent retrieval architecture that combines semantic vector search and keywordbased retrieval to extract context from SEC 10K filings, annual disclosures, and structured Snowflake databases, ensuring both semantic depth and lexical precision.

2.Teacher–Student Distillation: Deployment of a highcapacity teacher model for deep table and text understanding, guiding a lightweight student model to replicate extraction and reasoning with optimized efficiency, enabling faster tabular comprehension and lower inference cost.

3.EvaluatorintheLoop Systems: Integration of an evaluator metaagent that assesses responses for factual accuracy, faithfulness, and numerical consistency, automatically triggering query refinement or reranking cycles when confidence scores fall below the configured threshold, thus creating a continuous selfcorrection pipeline.

4.CrossModel Verification: Adoption of a verification layer using multiple large language models operating in parallel to crossverify results; inconsistencies are resolved through consensus scoring, ensuring that final outputs remain grounded, validated, and free from hallucination.

5.Financial Data Understanding: Application of AI models on structured and unstructured financial data—including SEC 10K reports, income statements, cashflow summaries, and risk disclosures—to achieve accurate extraction of financial entities, trend patterns, and key performance metrics through domainadapted embeddings.

6.Software Engineering and Deployment: Development of a containerized, modular, and configurationdriven system using Docker, FastAPI, and Streamlit, enabling secure, onpremise installations compliant with enterprise datagovernance standards; the design supports easy customization and scalability via YAMLbased configuration management.

2. Background

Modern financial analysis depends heavily on the ability to extract precise insights from dense, multimodal data sources such as SEC 10K filings, annual reports, investor briefings, and regulatory disclosures. These documents often exceed hundreds of pages, filled with domainspecific terminology, complex tables, and contextual dependencies that are not easily captured by generic natural language models. The volume and velocity of such data have increased exponentially, making manual exploration both timeconsuming and errorprone.

Traditional RetrievalAugmented Generation (RAG) platforms and generalpurpose AI assistants (such as cloudhosted LLMs) provide partial support for financial data retrieval, but their deployment in enterprise contexts encounters several constraints. They often:

- Generate hallucinated or unverifiable results due to insufficient crosschecking;
- Fail to respect dataprivacy mandates, as enterprise and financial content cannot be shared outside secured networks;
- Lack domainadapted embeddings and retrieval logic capable of interpreting financial labels, numerical tables, or accounting context; and
- Operate as static singleretriever pipelines that cannot identify or correct their own errors.

At Voya Global Services Pvt Ltd, analysts regularly deal with fragmented data ecosystems integrating structured repositories (e.g., Snowflake data warehouses, relational ledgers) and unstructured sources (e.g., SEC filings, audit documents, analyst commentaries). Retrieving an answer to a simple query—such as analyzing yearonyear revenue variance or linking risk disclosures to numerical tables—requires manually querying several systems, performing repeated reconciliation, and validating against corporate data warehouses. This workflow creates latency, redundancy, and inconsistent results across teams, weakening decisionmaking efficiency.

To address these operational and technical pain points, this dissertation proposes a SelfCorrecting Financial Intelligence System, a complete endtoend framework built on the principles of agent orchestration, evaluative feedback, and interpretabilitydriven AI. The system introduces a coordinated set of intelligent agents that collaborate and critique one another within a unified workflow, overcoming the shortcomings of conventional retrieval architectures.

The proposed framework is centered around five synergistic design components:

- LangGraphbased orchestration: A graphstructured control layer that coordinates retrieval, reasoning, and verification among autonomous agents, enabling dynamic task routing and scalable workflows.

- Hybrid RAG pipeline (FAISS + BM25): A dualretrieval mechanism combining semantic embedding search with lexical keyword matching to achieve both contextual understanding and exact factual retrieval.
- Evaluatorcontrolled feedback loop: An evaluator agent continuously evaluates outputs for faithfulness and relevance, initiating automatic selfcorrection cycles when responses lack confidence or factual grounding.
- Teacher–Student Distillation: A dualmodel learning strategy where a large teacher model performs accurate table and text reasoning, transferring structured knowledge to a lighter student model optimized for realtime performance on enterprise hardware.
- CrossModel Verification (Triangulation): Multiple large language models run in parallel to generate and validate candidate responses; discrepancies trigger consensus validation to eliminate hallucination and enhance trustworthiness.

The entire system is implemented as an onpremise, containerized solution to meet stringent enterprise security and regulatory requirements. Packaged using Docker and managed via a single configuration file, the solution ensures portability and ease of deployment across financial departments without exposing sensitive data to external cloud services. Its end goal is to deliver a transparent, explainable, and auditable AIbased assistant capable of producing consistent, datagrounded, and repeatable financial insights suitable for enterprise adoption and decision support.

3. Objectives

The central goal of this dissertation is to conceptualize, design, and implement a SelfCorrecting Financial Intelligence System that combines the strengths of RetrievalAugmented Generation (RAG), teacher–student distillation, evaluatordriven feedback loops, and multiagent collaboration to achieve verifiable, contextaware financial insight generation. The system aims to set new standards for accuracy, interpretability, and autonomy in financial analytics powered by artificial intelligence.
The specific objectives are outlined below:

1. To architect a LangGraphbased multiagent RAG framework that enables dynamic coordination among Retriever, Reader, Verifier, and Evaluator agents, ensuring seamless information flow, efficient task routing, and adaptive feedback control throughout the retrieval and reasoning process.
2. To implement a Teacher–Student Distillation pipeline that transfers indepth tabular reasoning and text understanding capabilities from a large teacher model to a smaller student model, optimizing performance for realtime, onpremise financial intelligence queries.
3. To develop a hybrid retrieval layer that integrates dense vector similarity search (FAISS) and lexical BM25 matching under a unified scoring mechanism, suitable for extracting information from both textual narratives and numerical table structures in financial reports.
4. To construct an Evaluator Agent capable of intelligent selfcorrection, applying multicriteria scoring frameworks—relevance, factual consistency, entailment, and numerical coherence—to automatically trigger query reformulation and iterative reretrieval when accuracy thresholds are not met.
5. To design a CrossModel Verification engine that performs consensus validation among multiple LLMs, minimizing hallucination and model bias by reconciling divergent outputs through automated scoring and majority agreement principles.
6. To integrate financial data from structured and unstructured sources, combining Snowflake data warehouse queries with content from SEC 10K filings, annual reports, and investor disclosures, thus creating a holistic knowledge corpus for retrieval and analytics.
7. To create a scalable backend implementation using FastAPI that exposes retrieval, reasoning, and evaluation modules through secure REST APIs, supporting asynchronous processing and enterprisegrade integrations.
8. To design an interactive Streamlitbased user interface that allows analysts to input naturallanguage queries, visualize retrieved evidence, inspect answer confidence scores, and trace model reasoning paths, thereby improving system explainability and user trust.

9.To ensure data privacy and compliance by deploying the entire framework in a containerized Docker environment, with configurable YAMLdriven setup that permits rapid adaptation to various enterprise infrastructures without external data transmission.

10.To perform finetuning and domain adaptation of embedding models using organizationspecific financial datasets, ensuring accurate representation of domain entities such as revenue segments, financial ratios, risk labels, and accounting terminology.

11.To establish automated logging, monitoring, and modelperformance tracking mechanisms, capturing operational metrics, error patterns, and evaluator confidence distributions for continuous model improvement.

12.To evaluate system performance across retrieval and reasoning layers using both quantitative and qualitative benchmarks,
including Precision@K, Mean Reciprocal Rank (MRR), Normalized Discounted Cumulative Gain (NDCG), Faithfulness Score, HallucinationDetection Accuracy, Numerical Consistency,
and CrossModel Agreement Rate.

13.To conduct comparative ablation studies assessing the contribution of individual modules—such as evaluator feedback, distillation, and hybrid retrieval—to the overall system performance and reliability.

14.To validate deployability and scalability on standard computing hardware, confirming that the solution can operate as a standalone, locally deployable assistant within enterprise networks without dependency on external LLM APIs.

Together, these objectives collectively ensure the development of a robust, interpretable, and enterprisegrade multiagent AI system that bridges the gap between academic innovation and practical financial data intelligence by offering transparency, verification, and selfimproving knowledge retrieval capabilities.

4. Scope of Work

The scope of this dissertation encompasses the endtoend design, development, integration, and evaluation of an advanced multiagent Financial Intelligence System built on RetrievalAugmented Generation (RAG) principles. It defines the technical boundaries, deliverables, and exclusions within which the research will be executed, ensuring precision, reproducibility, and alignment with M.Tech research standards.

The project covers the following core components:

1.***Comprehensive System Architecture and Modular AI Implementation:***

The system will be designed as a fully modular, Pythonbased framework implementing a clear separation of concerns among retrieval, reasoning, verification, and orchestration layers. Each component—Retriever, Reader, Verifier, Evaluator, and Orchestrator—will be independently configurable and reusable across multiple use cases.

2.***Integration of MultiAgent Workflow Using LangGraph Orchestration:***

Multiple specialized agents such as the Retriever, Reader, Verifier, and Evaluator will operate in parallel and communicate through a LangGraphbased workflow engine. This ensures efficient coordination, context propagation, and dynamic task routing, allowing the system to retrieve, analyze, and verify financial data autonomously.

3.***Teacher–Student Distillation and DomainSpecific Training:***

A Teacher–Student training module will be developed using structured financial table extraction datasets curated from SEC 10K reports and balance sheets. The large teacher model will supervise a compact student model to accurately reproduce tabular understanding and numerical reasoning with reduced computational overhead.

4.***SelfCorrecting Evaluator Feedback Mechanism:***

The Evaluator Agent will continuously monitor the system's responses for factual consistency, numerical accuracy, and semantic relevance. When confidence metrics drop below defined thresholds, it will automatically trigger query expansion, reretrieval, or passage reranking to refine the results—enabling a fully autonomous selfcorrection cycle.

5.***Hybrid Retrieval Integration (FAISS + BM25):***

The retrieval subsystem will employ a hybrid approach combining dense vector search using FAISS and sparse keyword search using BM25. A weighted scoring strategy will balance semantic similarity and exact keyword matching to achieve optimal retrieval precision across textual narratives and embedded financial tables.

6.***Snowflake Data Warehouse Integration:***

The system will connect natively with the Snowflake data warehouse to enable highperformance querying of structured financial data—such as revenue, expenses, and risk indicators—alongside unstructured textual content from regulatory filings.

7.***BackEnd and User Interface Development:***

A FastAPI backend will expose retriever and evaluator endpoints as RESTful services, while a Streamlit web frontend will provide users with query inputs, confidencescore visualization, supporting evidence display, and citation transparency for every AIgenerated result.

8.***Containerized Deployment and Configuration Management:***

The entire system will be packaged as a Dockerized application supporting singlecommand installations. Configuration will be managed through a YAMLbased orchestrator file that defines data sources, model paths, and confidence thresholds. This modular setup facilitates privacycentric, onpremise deployment with minimal user intervention.

9.***Evaluation and Benchmarking of System Performance:***

Rigorous empirical studies will be conducted to measure componentlevel and endtoend system effectiveness. Evaluation metrics will include Precision@K, Mean Reciprocal Rank (MRR), Normalized Discounted Cumulative Gain (NDCG), Faithfulness Score, and Evaluator Hit Rate. Additional ablation experiments will quantify the impact of individual modules such as evaluator feedback and hybrid retrieval.

10.***Version Control and Documentation:***

The system will maintain full version control using Git for traceability and reproducibility. Detailed documentation covering architecture design, configuration settings, data schema, and deployment steps will be developed for academic and industrial audiences.

Explicitly Excluded from Scope:

The dissertation will not involve:

- Training or developing large foundational language models (LLMs) from scratch;
- Integration with proprietary or paid cloudbased APIs (e.g., OpenAI, Anthropic, Gemini) due to dataprivacy and cost constraints;
- Largescale enterprise rollouts or production hosting beyond the demonstration environment;
- Handling or processing live financial transactions or sensitive data outside the approved research environment.

This defined scope ensures that the dissertation remains technically rigorous, academically measurable, and practically executable, focusing on implementable innovations that improve trust, explainability, and automation in enterprisegrade financial knowledge retrieval.

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5. **Plan of Work**

Phase	Timeline	Work to be Done
Phase 1: Outline & Literature Review	25 Apr – 10 May 2026	Finalize scope, perform literature review on RAG, multi-agent orchestration, distillation, and evaluator-based correction; submit
Phase 2: Data Ingestion & Baseline RAG	11 May – 4 Jun 2026	Implement SEC ingestion (PDFium2), Snowflake integration, and basic retriever; design Streamlit UI skeleton.
Phase 3: Multi-Agent Architecture & Teacher–Student Distillation	5 Jun – 21 Jun 2026	Build orchestrator in LangGraph; teacher–student training for table interpretation; conduct mid-semester evaluation.
Phase 4: Verifier & Evaluator Feedback Loop	22 Jun – 15 Jul 2026	Add verifier agents and evaluator self-correction logic; integrate cross-model consistency checks.
Phase 5: Docker Packaging & Evaluation	16 Jul – 2 Aug 2026	Complete Docker containerization; run benchmark evaluations; submit final dissertation report.
Phase 6: Presentation and VIVA	3 Aug – 28 Aug 2026	Prepare evaluation dashboard; record demonstration; appear for final VIVA examination.

6. Literature References

A thorough understanding of existing stateoftheart research underpins the success of any project. The following publications have been selected through a preliminary literature survey as highly relevant to the proposed system, encompassing retrievalaugmented generation, embedding techniques, multiagent architectures, and evaluator methodologies.

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[20] Ribeiro, J. & Dogan, P., “FaithEval: Systematic Evaluation of Faithfulness Metrics for RAG,” NA ACL Workshops, 2025

7. Particulars of the Supervisor and Examiner

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8. Remarks of the Supervisor

The paragraph is essentially a supervisor's formal endorsement and assessment of the dissertation proposal by Karthik V, highlighting its technical merit, research depth, and practical significance. Here's a detailed explanation of its meaning and implications:

1.Technically strong and practically relevant initiative addressing a critical need in enterprise financial analytics.

1. This means the proposal is not only welldesigned from an engineering and AI standpoint but also solves a realworld problem faced by businesses—particularly, how to make financial analysis more intelligent, accurate, and automated. It balances academic research with practical industrial application.

2. **System design—featuring multi-agent retrieval, teacher–student distillation, cross-model verification, and evaluator-driven self-correction—demonstrates both academic rigor and industrial applicability.**

   1.The supervisor recognizes that the system is composed of advanced and wellresearched technical components.

   1. *Multi-agent retrieval* enables distributed information gathering.
   2. *Teacher–student distillation* improves learning efficiency.
   3. *Cross-model verification* ensures factual reliability by comparing outputs from multiple AI models.
   4. *Evaluator-driven self-correction* gives the system the ability to detect and fix its own inaccuracies.

   2. Together, these elements make the project scientifically rigorous (suitable for research) and immediately useful for real industrial scenarios (suitable for deployment).

3. **The student's approach combines established research from retrieval-augmented generation and self-correcting AI with an emphasis on trustworthy and explainable answers.**

   1. The work builds upon proven concepts in RAG and self-correcting AI systems, but extends them toward *trustworthy AI*, focusing on outputs that can be verified, explained, and justified—critical aspects in financial domains where data accuracy is nonnegotiable.

4. **The design ensures on-premise, compliant operation suitable for financial ecosystems.**

   1. The solution is developed to run within an organization's infrastructure (onpremise), protecting sensitive financial data and maintaining compliance with informationsecurity and datagovernance regulations. This aligns perfectly with how real financial institutions handle confidential data.

5. **Given the structured plan, achievable milestones, and clear deliverables**

   1. The supervisor confirms that the project timeline, goals, and implementation steps are welldefined and realistic. The student demonstrates a clear understanding of how to complete the research successfully within the academic schedule.

6. **Meets the academic depth required at the M.Tech level and holds potential for publication-grade results and industrial deployment**

    1. The project is of sufficient intellectual and technical depth to fulfill the requirements of a Master's dissertation. Moreover, the quality and novelty of the work could lead to academic publications or realworld adoption within the financial industry.

7. **The proposal is approved and supervision will be provided throughout all research phases.**
    1. This is the official validation that the supervisor endorses the proposal. It means the project has been accepted as an M.Tech dissertation topic, and the student will receive continuous mentorship and evaluation during development, experimentation, and documentation stages.

In summary, the section expresses that Karthik V's dissertation is welljustified, technologically advanced, academically rigorous, and practically deployable, and that the supervisor fully supports and approves its execution and evaluation.

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Signature of Student	**Signature of Supervisor**	**Signature of Additional Examiner**
Name: Karthik V	Name: ShivaChandran Masilamani	Name: Deepthi B